

Topic B5

Enzymes

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B5.1	Enzyme action
B5.2	Effects of temperature and pH
B5.3	Investigating enzymes

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Enzyme action

Enzymes are proteins that catalyse metabolic reactions. A substrate with a complementary shape binds at the active site to form an enzyme-substrate complex; products leave and the enzyme can be reused. Enzymes lower activation energy.

Examiner insight

Use “complementary” rather than saying the substrate is the same shape as the active site.

2. Temperature

Raising temperature initially increases kinetic energy and successful collisions. Above the optimum, bonds maintaining enzyme shape break, the active site changes and the enzyme denatures; rate falls rapidly. Low temperature slows reactions but does not normally denature.

Examiner insight

Explain both sides of a temperature-rate graph.

3. pH

Changes in pH alter ionic interactions and active-site shape. Each enzyme has an optimum pH related to its location and function; extreme pH can denature it. Pepsin works in acidic stomach conditions, while many intestinal enzymes prefer alkaline conditions.

Examiner insight

Avoid claiming all enzymes have optimum pH 7.

4. Experimental design

Rate can be measured by product formation or substrate disappearance. Keep enzyme and substrate concentrations, volumes and temperature controlled when testing pH. Use buffers, repeats and an objective end point such as a colorimeter.

Examiner insight

Rate should be calculated from an initial gradient or reciprocal time when appropriate.

Worked application

Situation: A patient has a fever well above an enzyme optimum.

Prediction: Reaction rate may decrease.

Reason: High temperature disrupts bonds, changes the active site and reduces enzyme-substrate complexes.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B5

Enzymes**Instructions**

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines enzyme? [1]
 - A a protein that acts as a biological catalyst
 - B the region of an enzyme where the substrate binds
 - C the reactant molecule on which an enzyme acts
 - D a permanent change in protein shape that alters the active site
2. Which statement best defines active site? [1]
 - A the reactant molecule on which an enzyme acts
 - B the region of an enzyme where the substrate binds
 - C a permanent change in protein shape that alters the active site
 - D the condition at which enzyme rate is greatest
3. Which statement best defines substrate? [1]
 - A a permanent change in protein shape that alters the active site
 - B the condition at which enzyme rate is greatest
 - C the reactant molecule on which an enzyme acts
 - D a protein that acts as a biological catalyst
4. Which statement best defines denaturation? [1]
 - A the condition at which enzyme rate is greatest
 - B a protein that acts as a biological catalyst
 - C the region of an enzyme where the substrate binds
 - D a permanent change in protein shape that alters the active site
5. Which statement best defines optimum? [1]
 - A the condition at which enzyme rate is greatest
 - B a protein that acts as a biological catalyst
 - C the region of an enzyme where the substrate binds
 - D the reactant molecule on which an enzyme acts
6. Which statement about enzyme action is correct? [1]
 - A A reactant used up in metabolism.
 - B Enzymes are proteins that catalyse metabolic reactions.
 - C The part of a cell where dna is stored.
 - D The final product of a reaction.
7. Which statement about temperature is correct? [1]
 - A The part of a cell where dna is stored.
 - B The final product of a reaction.
 - C Raising temperature initially increases kinetic energy and successful collisions.
 - D A reversible slowing caused by low temperature.

8. Which statement about pH is correct? [1]
- A The final product of a reaction.
 - B A reversible slowing caused by low temperature.
 - C The highest condition tested regardless of rate.
 - D Changes in pH alter ionic interactions and active-site shape.
9. Which statement about experimental design is correct? [1]
- A Rate can be measured by product formation or substrate disappearance.
 - B A reversible slowing caused by low temperature.
 - C The highest condition tested regardless of rate.
 - D A reactant used up in metabolism.
10. Which prediction is most appropriate when a patient has a fever well above an enzyme optimum? [1]
- A A reactant used up in metabolism.
 - B The final product of a reaction.
 - C Reaction rate may decrease.
 - D The highest condition tested regardless of rate.

Section B · Structured questions

11 Biologists use precise vocabulary when studying enzymes.

(a) Define enzyme.

[2]

(b) Define active site.

[2]

(c) Distinguish substrate from the incorrect idea: “the final product of a reaction”.

[2]

(d) State two important features of enzyme action.

[2]

[Total: 8]

12 A patient has a fever well above an enzyme optimum.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative rate under three conditions.

condition	relative rate
20 °C	2
40 °C	9
70 °C	0

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to investigate pH effect on amylase breaking down starch.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of enzymes to an unfamiliar context.

(a) Define denaturation. [2]

(b) Explain the biological importance of pH. [3]

(c) Correct this statement: “the highest condition tested regardless of rate”. [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A protein that acts as a biological catalyst.
- 2 B. The region of an enzyme where the substrate binds.
- 3 C. The reactant molecule on which an enzyme acts.
- 4 D. A permanent change in protein shape that alters the active site.
- 5 A. The condition at which enzyme rate is greatest.
- 6 B. Use “complementary” rather than saying the substrate is the same shape as the active site.
- 7 C. Explain both sides of a temperature-rate graph.
- 8 D. Avoid claiming all enzymes have optimum pH 7.
- 9 A. Rate should be calculated from an initial gradient or reciprocal time when appropriate.
- 10 C. High temperature disrupts bonds, changes the active site and reduces enzyme-substrate complexes.

Solutions · Structured questions

11. (a) a protein that acts as a biological catalyst [2]. (b) the region of an enzyme where the substrate binds [2]. (c) the reactant molecule on which an enzyme acts; not the final product of a reaction [2]. (d) Credit any two accurate features from the guide: Enzymes are proteins that catalyse metabolic reactions. A substrate with a complementary shape binds at the active site to form an enzyme-substrate complex; products leave and the enzyme can be reused. Enzymes lower activation energy. [2].

12. (a) Reaction rate may decrease [2]. (b) High temperature disrupts bonds, changes the active site and reduces enzyme-substrate complexes [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $9 - 0 = 9$ [2]. (b) The values change from 2 to 0, with the highest value 9. [2]. (c) Rate rises toward an optimum then falls at high temperature because the active site denatures [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Buffer pH [1]. (b) Time until iodine remains orange-brown [1]. (c) Same temperature, enzyme and starch concentrations and volumes [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use a thermostatic water bath and repeat each pH [1].

15. (a) a permanent change in protein shape that alters the active site [2]. (b) Award linked points from: Changes in pH alter ionic interactions and active-site shape. Each enzyme has an optimum pH related to its location and function; extreme pH can denature it. Pepsin works in acidic stomach conditions, while many intestinal enzymes prefer alkaline conditions. [3]. (c) Correct statement: optimum means the condition at which enzyme rate is greatest [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.